

WHITE PAPER · NOVEMBER 2026

Who Decided?

Deterministic Safety for AI in Health Plan Utilization Management

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Models propose, deterministic rules decide, and licensed clinicians sign.

AI accountability belongs in the architecture of each determination. This paper describes a bounded role for probabilistic models, a deterministic core over versioned criteria, and a sealed record that makes the division of responsibility inspectable.

For health plan executives, utilization management leaders, compliance and legal teams, and architects.

Aejaz Sheriff Quaraishi
Architecture paper, not legal advice.

EXECUTIVE SUMMARY

Every AI determination must answer one question: who decided?

Appeals, audits and depositions need per-case evidence, not a policy statement about human oversight.

“Assistive tool, clinician makes the final call” leaves essential questions unanswered: which facts, criteria and policy version applied, and what evidence did the reviewer see, change and sign?

The Deterministic Safety Principle: models propose, deterministic rules decide, and licensed clinicians sign.

Four commitments make the answer inspectable

01

Bound the model’s job. Separate extraction, drafting and explanation from coverage authority.

02

Keep a deterministic core. Evaluate structured facts against versioned criteria using CQL over FHIR and DMN tables.

03

Never automate a denial. Unresolved and adverse outcomes require competent licensed clinical review.

04

Seal a replayable trace. Preserve evidence, evaluation, routing and human actions in one record.

For health plan CIOs, CMOs, UM leaders, compliance and legal teams, and architects. This is an architecture paper, not legal advice.

EXTERNAL PRESSURE

Regulators and courts are converging on the same demand

Different authorities raise different questions, but each makes the boundary between machine contribution and human decision consequential.

Litigation case-level discovery	Estate of Lokken v. UnitedHealth concerns nH Predict; discovery opened March 2026. Kisting-Leung v. Cigna concerns PxDx; core claims proceeded in March 2025. Neither case is decided on the merits.
California SB 1120 clinician authority	Effective January 1, 2025; bars AI from adverse medical necessity determinations.
Texas SB 815 separation of roles	Applies to plans issued or renewed from January 1, 2026; bars AI from adverse medical necessity determinations.
CMS Medicare Advantage rule individual evidence	Requires individualized determinations and static, public criteria.
CMS-0057-F reasons, clocks and exchange	Specific denial reasons and 72-hour and 7-day clocks since January 1, 2026; FHIR Prior Authorization API by January 1, 2027.

↓

One per-determination record

Show where the machine's proposal ended and the licensed person's decision began.

EVIDENCE OBLIGATIONS

Read across the rows: each source asks for a record per determination

The practical response is a record that joins authority, evidence and action. A general AI policy cannot establish what happened in a particular case.

Source	Accountability question	What the plan must produce
CA SB 1120	Who exercised clinical authority?	Identity and competence of the deciding clinician; proof the tool did not set the outcome.
TX SB 815	Were automated steps separate from the adverse decision?	Clean role separation and audit-ready logs.
CMS MA rule and FAQ	Was this patient evaluated against the applicable criteria?	Patient-specific evidence and the exact criteria version applied.
CMS-0057-F	Does the reason match the basis, within the required time?	Reasons traceable to criteria and evidence; timestamps and path-level metrics.
Litigation discovery	What did the machine and reviewer each contribute?	Per-case model contributions, reviewer actions and override patterns.

Read the final column as a design specification

These artifacts should exist during ordinary operations. Reconstructing them after an appeal or discovery request leaves gaps precisely where the plan needs a defensible account.

FAILURE → REQUIREMENT

“AI-assisted” fails five tests, and each failure sets a design requirement

The label describes participation, not control. Each ambiguity needs an architectural answer that can be verified in a case record.

Failure		Design requirement
“Assist” hides three jobs: produce evidence, recommend and effectively decide.	→	Name each component’s job; make effective model decision-making impossible.
A signature alone does not establish meaningful human review.	→	Capture what the reviewer saw, changed and how long they spent.
Probabilistic outputs are not reproducible on demand.	→	Record model identity, configuration version and output hash at use.
Model drift can silently become criteria drift.	→	Keep criteria explicit and versioned; models may inform but never redefine them.
Automation bias can turn review into ratification.	→	Present evidence and criteria rather than the machine’s verdict.

The JAMIA review of 74 studies on automation bias supports treating human review as a workflow design problem. The workbench must support independent judgment, while the trace makes that judgment visible.

Human-factors reference: Goddard, Roudsari and Wyatt, JAMIA 2012.

THE GOVERNING PRINCIPLE

The Deterministic Safety Principle: models propose, rules decide, humans sign

Probabilistic model output never directly drives a coverage determination, a dollar figure or a clinical decision.

Propose →
MODELS

Candidate facts, drafts and explanations, each pointing to its source.

Decide →
RULES

Versioned criteria evaluated on structured data. Same inputs, same output.

Sign
LICENSED HUMANS

Competent clinicians resolve adverse or unresolved outcomes.

The model has no direct connection to a coverage outcome. Its proposals must pass validation and the deterministic core.

Separate calculation from clinical authority

The engine produces criterion results and a controlled route. It does not convert a failed criterion into an automated denial. The clinician sees the underlying evidence and applicable criteria before exercising judgment.

This division makes responsibility inspectable without claiming that rules capture every clinical situation. When the structured evaluation cannot resolve a case, the system assembles the evidence and routes it to review.

DESIGN CONSEQUENCES

Six corollaries turn the principle into design decisions

01

Model output is untrusted input

Validate schema, cited source span and plausibility before accepting a proposed fact.

02

Use a rule when there is one right answer

Use a model where many acceptable answers exist, such as a draft explanation.

03

Approval and denial are not symmetric

Where law allows, automation may route toward approval, but never deny, delay or modify.

04

Fail toward review

Missing, conflicting or unevaluable evidence goes to human review, never default denial.

05

Criteria are code

Link each encoded criterion to a clause of the published medical policy.

06

Keep the determination pipeline fixed

No agentic planning in the determination path. Every possible route is known in advance.

The principle does not exclude AI, claim that rules capture all clinical judgment, or treat a signature as proof of a real decision. It limits where uncertainty can acquire decision authority.

REFERENCE ARCHITECTURE

One invariant shapes the reference architecture

No path from model output to a determination bypasses the deterministic core; no path to an adverse determination bypasses a licensed clinician's signature.

Intake
X12 278 · Da Vinci CRD/DTR/PAS over FHIR · portals · faxed documents

Neural accelerators
Note extractor
Rule authoring assistant
Explanation generator
Reviewer copilot

Governed evidence
Structured source data
Human-confirmed facts
Immutable policy snapshot
Signed rule releases

↓ Neural-symbolic boundary ↓
Only validated, source-linked candidate facts cross

Deterministic core
CQL over FHIR + DMN tables
Each criterion: met · not met · not evaluable

↓ Routing gate ↓
No denial route

Approval-eligible	Clinician review	Request for information	Pend

Clinician workbench
Evidence + criteria → independent judgment → licensed signature for adverse outcomes

Sealed trace across every layer
SHA-256 hash-chained record

AI control plane
Model gateway with BAA enforcement · signed prompt registry · guardrails · observability

Authoring support and read-only assistance remain outside runtime decision authority. Routing and timing controls remain visible even when evidence is incomplete.

BOUNDARY CONTRACT

Only source-linked structured facts may cross the neural-symbolic boundary

The core consumes defined fields, not persuasive language. A proposed fact must retain its source so the reviewer can inspect the evidence behind it.

Allowed through the gate	Stopped at the gate
Structured facts containing code, value, unit and date, with a source span.	Free text that asks the deterministic core to interpret a narrative.
Human-confirmed facts with their provenance preserved.	A model confidence score used as a decision threshold.
FHIR and X12 data in the defined input contract.	A model's recommended coverage outcome.
Unconfirmed extracted facts toward approval routing only when grounding passes and plan policy permits.	Any value without a source.

Confirmation is a policy choice, not a hidden default

A conservative plan confirms every extracted fact. A plan permitting unconfirmed facts toward approval must make that choice explicit and preserve the grounding result. Neither approach gives model output authority to produce an adverse determination.

This contract contains model risk and supports reproducible evaluation. It creates the supporting evidence before an appeal, audit or deposition asks for it.

BOUNDED MODEL ROLES

Four bounded roles let plans deploy imperfect models responsibly

Each accelerator has a useful output, a prohibited action and a control matched to its failure mode.

Role / proposes	Can never	Failure mode	Control
Note extractor Facts with source spans.	Judge whether criteria are met.	Unsupported facts, wrong units or implausible values.	Span grounding, unit and range rules, human confirmation.
Rule authoring assistant CQL, DMN and tests.	Deploy criteria into production.	A draft encodes the wrong policy meaning.	Clinical and technical review; signed release.
Explanation generator Notices from the trace only.	Add a new reason for the outcome.	The notice diverges from the actual basis.	Trace-bound drafting; reviewer edits retained.
Reviewer copilot Read-only cited answers.	Write to the determination or trace.	Unsupported guidance influences review.	Read-only access and source-linked answers.

Contain the consequence of error

The architecture assumes models can be wrong. Responsibility comes from constraining what an error can change, validating each output and preserving the contribution. Better model quality remains valuable, but it never substitutes for a boundary on authority.

THE EVIDENCE PACKAGE

The evidence package: the life of one determination record

One case accumulates eight linked elements. Each entry retains its connection to the prior record rather than replacing the history.

1 **Request → evidence set**

Preserve the request and the evidence available for evaluation, including source-linked facts.

2 **Policy snapshot → rule evaluation**

Bind the exact criteria version to each met, not met or not evaluable result.

3 **Model contributions → route**

Retain what each model proposed and the controlled route selected by the gate.

4 **Reviewer actions → notice**

Capture review and signature; link the notice to the actual evaluation basis.

Hash-linked sequence: request → evidence → evaluation → routing → clinical review → notice. Policy snapshots and model contributions attach at their points of use.



Make the human contribution visible

Record identity, credentials, specialty, what was viewed, time on case, edits and signature. The notice must state the reason supported by the rule evaluation, not a newly generated rationale.

INTEGRITY AND INDEPENDENT REVIEW

Sealing, replay and monitors keep review from becoming ratification

Record integrity and meaningful judgment are different problems. The system needs controls for both.

Seal → preserve the history

Write-once entries are hash-chained and periodically anchored to a store operators cannot modify. The chain makes alteration detectable; the protected anchor supports the integrity claim.

Replay → verify the evaluation

Recorded evidence and policy snapshot reproduce the same criterion results. Appeals can verify the evaluation, and policy changes can be tested against past cases.

Monitor → inspect the human process

Track override and correction rates, time on case by route, appeal overturn rates by route, and drift-right changes in the routing mix.

Edits are evidence, not noise

Reviewer corrections and changes are signs of independent judgment. Fast signatures alone do not prove deficient review, nor does long dwell time prove a good one. Read these signals alongside what was viewed, changed and decided.

Replay establishes computational consistency, not that the policy was clinically sound or the reviewer exercised judgment.

CONTROL → ARTIFACT

Regulatory mapping: each requirement has a control and an artifact

A shared evidence set can support several obligations. The mapping below is an architectural response, not a claim that a technical control alone establishes compliance.

Requirement	Architectural control	Evidence artifact
CA SB 1120 / TX SB 815	No denial route; licensed clinician signature; evidence-first workbench.	Route record, signature and workbench view log.
CMS MA rule	Patient-specific evaluation; criteria linked to posted policy.	Evidence set, immutable policy snapshot and criterion results.
CMS-0057-F	Reasons from evaluation; timed routes; FHIR-native intake.	Reason-to-evidence links, timestamps and path metrics.
Cross-cutting governance	Controlled access, model inventory, evaluation gates and monitoring.	Model contributions, releases, review history and audit records.

Reuse evidence without collapsing legal distinctions

Section 1557, HIPAA, the NAIC Model Bulletin, Colorado SB 26-189, NIST AI RMF and ISO/IEC 42001 can draw on this common record. Their scopes and requirements remain distinct.

Colorado SB 26-189 is identified here as effective January 1, 2027 and rulemaking-dependent. Legal and compliance teams must evaluate applicability and implementation details.

EVALUATION DESIGN

Baseline before benefit: measure speed, accuracy and clinician time

Capture the human process before any model goes live. Pre-register what improvement will count as success, then attribute results to the route that produced them.

Optimize service and quality	Track, but never target
Turnaround time and notice defects.	Denial rate.
Reviewer time, split into finding evidence and judging the case.	Avoided spend.
RFI rate and the share clearly meeting criteria.	Outcome mix used as a financial performance goal.
Appeals and overturns as quality signals.	Any incentive that rewards producing more adverse outcomes.

Attribute by route

Approval-eligible · clinician review · request for information · pend

Compare each path with its baseline. An aggregate improvement can conceal extra work or poorer outcomes in another path.

Count the full operating cost

Include rule encoding, subject-matter expertise and governance effort. Time saved finding evidence is different from time saved exercising judgment. Measure them separately so a faster workflow does not quietly become a shallower review.

ADOPTION ROADMAP

A five-phase adoption path, each phase gated by an exit criterion

Advance on evidence of readiness, not on a calendar commitment. Each phase preserves the prohibition on an automated denial.

01	02	03	04	05
Map	**Instrument**	**Build**	**Add AI**	**Route**

Phase	Exit criterion
Map the boundary	Written map of every point where machine output can reach a determination.
Instrument human review	Sealed manual-review records exist and the baseline is measured.
Build the core	Deterministic evaluation has been exercised in shadow mode before live use.
Add bounded accelerators	Each remains behind the gate and is evaluated against pre-registered targets.
Enable approval routing	Criteria are clear and law allows the route. No denial route is enabled.

Management system beneath every phase

AI inventory · risk assessment · evaluation gates · monitoring · audit

Require vendors to export per-case contributions, separate extraction from recommendation and share evaluation results. A vendor boundary must not become a gap in the plan’s determination record.

OBJECTIONS AND LIMITS

Objections and limits: where the approach runs out

Rules cannot capture all clinical judgment	They handle the decidable part and route the rest to clinicians with evidence assembled. A not evaluable result is a reason for review, not a proxy for denial.
Encoding is costly	Start with well-specified categories, reuse value sets and use an assistant to draft rules and tests. Clinical and technical review remain necessary before release.
Deterministic does not mean correct	A wrong rule can reproduce a wrong answer consistently. Explicit rules make the error visible, fixable and replayable; they do not guarantee sound medical policy.
Human review can still become rubber-stamping	Evidence-first presentation and review logs reduce that risk and make it more visible. Neither a signature nor a vendor's assurance proves independent judgment. Vendors do not transfer the plan's responsibility.

The boundary of the claim

This approach cannot repair poor medical policy, settle legal questions or govern every AI use. It depends on people who maintain criteria, investigate signals and exercise clinical authority when the pipeline reaches its limits.

AUTHOR AND SOURCE REGISTER

About the author and references

Aejaz Sheriff Quaraishi is an enterprise and AI architect with nearly three decades in healthcare payer platforms. Views are his own; nothing here is legal advice.

References by subject

1. Litigation: *Estate of Lokken v. UnitedHealth*; *Kisting-Leung v. Cigna*. Procedural developments cited in this paper are not decisions on the merits.
2. State law: California SB 1120; Texas SB 815; Colorado SB 26-189. Applicability and rulemaking details require legal review.
3. Federal and insurance context: CMS Medicare Advantage FAQ, February 2024; CMS-0057-F; NAIC survey and Model Bulletin.
4. Human factors: Goddard, Roudsari and Wyatt, JAMIA 2012, review of automation bias.
5. Governance standards: NIST AI RMF 1.0; ISO/IEC 42001:2023.
6. Implementation standards: HL7 Clinical Quality Language (CQL); OMG Decision Model and Notation (DMN); Da Vinci CRD, DTR and PAS.

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Source register reflects the materials identified for this paper. Legal obligations, procedural status and implementation guidance should be checked against current primary sources before deployment.